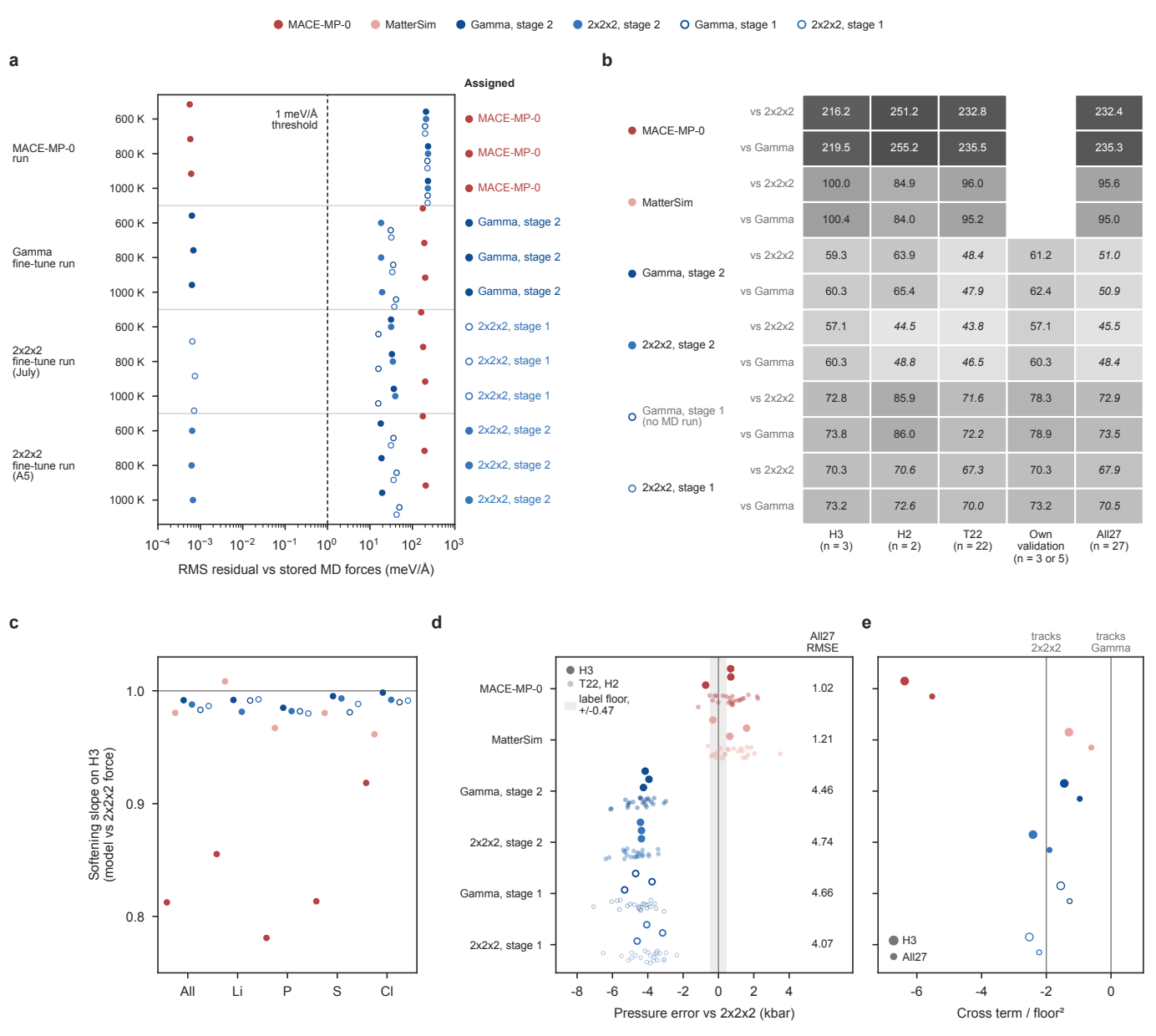

MACE-MP-0
MatterSim
Gamma, stage 2
2x2x2, stage 2
Gamma, stage 1
2x2x2, stage 1
a
Assigned
1 meV/Å threshold
MACE-MP-0 run
Gamma fine-tune run
2x2x2 fine-tune run (July)
2x2x2 fine-tune run (A5)
600 K
800 K
1000 K
RMS residual vs stored MD forces (meV/Å)
b
vs 2x2x2
vs Gamma
216.2 251.2 232.8 232.4
219.5 255.2 235.5 235.3
100.0 84.9 96.0 95.6
100.4 84.0 95.2 95.0
59.3 63.9 48.4 61.2 51.0
60.3 65.4 47.9 62.4 50.9
57.1 44.5 43.8 57.1 45.5
60.3 48.8 46.5 60.3 48.4
72.8 85.9 71.6 78.3 72.9
73.8 86.0 72.2 78.9 73.5
70.3 70.6 67.3 70.3 67.9
73.2 72.6 70.0 73.2 70.5
H3 (n = 3)
H2 (n = 2)
T22 (n = 22)
Own validation (n = 3 or 5)
All27 (n = 27)
c
Softening slope on H3 (model vs 2x2x2 force)
All
Li
P
S
Cl
d
All27 RMSE
H3
T22, H2
label floor, +/-0.47
MACE-MP-0 1.02
MatterSim 1.21
Gamma, stage 2 4.46
2x2x2, stage 2 4.74
Gamma, stage 1 4.66
2x2x2, stage 1 4.07
Pressure error vs 2x2x2 (kbar)
e
tracks 2x2x2
tracks Gamma
H3
All27
Cross term / floor²